A certified outer bound for convex NLP scenario subproblems

mpi-sppy

companion to `doc/designs/ipopt_outer_bound_design.md`

Abstract

The usual device for getting outer bounds in mpi-sppy is to read the dual bound off the subproblem solver, which fails for Ipopt, which reports no dual bound. This note derives the bound that the `ipopt_outer_bound` spoke computes instead. The construction is Lagrangian weak duality combined with a tangent-plane underestimator minimised in closed form over the variable box. *None of the mathematics is new*: Theorem 2 is the Frank–Wolfe duality gap on a box, and Section 8 places each result against the literature. The derivation is written out because the hypotheses are what matter in practice, and one of them is easy to get backwards. Its defining property is that it requires *no* assumption about the accuracy of the solve: a truncated or sloppy solve yields a loose bound rather than an invalid one, and an ill-conditioned one does the same, since the construction evaluates rather than solves and so admits no amplification of error by a condition number. Convexity, by contrast, is genuinely load-bearing, and Section 5 isolates the one place where it is easy to get backwards.

1 Setting

We assume minimization. Fix a scenario s with probability p_s, $\sum_s p_s = 1$. Let $v \in \mathbb{R}^n$ collect *all* variables of the scenario subproblem — the nonanticipative variables $x(v)$ together with the recourse variables — and let

$$B \;=\; \prod_{i=1}^{n} [\, l_i, u_i \,]$$

be the box of variable bounds. With hub weights W_s, the subproblem an outer bound spoke solves is

$$L_s(W_s) \;=\; \min_v \left\{ f_s(v) + W_s^\top x(v) \;:\; g_s(v) \le 0,\; h_s(v) = 0,\; v \in B \right\}, \tag{1}$$

with $f_s : \mathbb{R}^n \to \mathbb{R}$, $g_s : \mathbb{R}^n \to \mathbb{R}^m$ and $h_s : \mathbb{R}^n \to \mathbb{R}^k$. This is exactly the problem the ordinary Lagrangian spoke builds, with the proximal term switched off.

Why the returned objective value is not a bound here. For a minimisation, the value the solver reports is the objective evaluated at the point it stopped at. Any such value is $\ge$ the minimum, so it is an *inner* bound: the wrong direction. Nor does Ipopt's convergence tolerance close the gap, since it is a relative KKT-residual tolerance rather than a bound on the objective error, and the two error sources point opposite ways — optimality error stops slightly above the optimum, while constraint violation can place the iterate slightly below it.

2 Weak duality

Lemma 1 (Weak duality)**.** *Let $\lambda \in \mathbb{R}^m_+$ and $\mu \in \mathbb{R}^k$ be arbitrary. Define*

$$\varphi_s(v) \;=\; f_s(v) + W_s^\top x(v) + \lambda^\top g_s(v) + \mu^\top h_s(v), \tag{2}$$
$$q_s(\lambda, \mu) \;=\; \inf_{v \in B} \varphi_s(v). \tag{3}$$

Then $q_s(\lambda, \mu) \le L_s(W_s)$.

Proof. Let v be any point feasible for (1). Then $v \in B$, so $q_s(\lambda, \mu) \le \varphi_s(v)$. Moreover $g_s(v) \le 0$ and $\lambda \ge 0$ give $\lambda^\top g_s(v) \le 0$, and $h_s(v) = 0$ gives $\mu^\top h_s(v) = 0$; hence $\varphi_s(v) \le f_s(v) + W_s^\top x(v)$. Chaining the two inequalities, $q_s(\lambda, \mu) \le f_s(v) + W_s^\top x(v)$ for every feasible v. Taking the infimum over feasible v gives the claim. □

Note what Lemma 1 does *not* require: λ and μ need not be optimal, or even good. Any $\lambda \ge 0$ and any μ will do. This is the property that makes the whole approach robust, and it is why a mistaken sign convention or a stale multiplier can only cost tightness (Remark 4).

The trap. Lemma 1 is not yet usable, because q_s is defined by an *infimum*. Handing φ_s to a nonlinear solver returns a *point*, and the value there is $\ge$ the infimum — an upper bound on q_s, which is the wrong direction again. A second NLP solve therefore does not by itself certify anything.

3 The box underestimator

Theorem 2 (Certified bound)**.** *Suppose φ_s is convex on an open set containing B and differentiable at $\hat{v}$. Define*

$$\hat{q}_s(\hat{v}) \;=\; \varphi_s(\hat{v}) \;+\; \sum_{i=1}^{n} \min_{t \in [l_i, u_i]} \partial_i \varphi_s(\hat{v}) \left(t - \hat{v}_i\right). \tag{4}$$

Then $\hat{q}_s(\hat{v}) \le q_s(\lambda, \mu) \le L_s(W_s)$, and each term of the sum is available in closed form:

$$\min_{t \in [l_i, u_i]} \partial_i \varphi_s(\hat{v})\,(t - \hat{v}_i) \;=\; \begin{cases} \partial_i \varphi_s(\hat{v})\,(l_i - \hat{v}_i), & \partial_i \varphi_s(\hat{v}) > 0, \\ \partial_i \varphi_s(\hat{v})\,(u_i - \hat{v}_i), & \partial_i \varphi_s(\hat{v}) < 0, \\ 0, & \partial_i \varphi_s(\hat{v}) = 0. \end{cases} \tag{5}$$

Proof. Convexity and differentiability at $\hat{v}$ give the gradient inequality

$$\varphi_s(v) \;\ge\; \varphi_s(\hat{v}) + \nabla \varphi_s(\hat{v})^\top (v - \hat{v}) \qquad \text{for all } v.$$

Minimising both sides over $v \in B$ preserves the inequality, and the right-hand side is separable across coordinates because B is a box, which yields (4). Each one-dimensional problem minimises a linear function over an interval, so the minimum is attained at an endpoint, giving (5). The final inequality is Lemma 1. □

Corollary 3. *$\hat{q}_s(\hat{v}) \le L_s(W_s)$ for* any *$\hat{v}$ at which φ_s is differentiable,* any *$\lambda \ge 0$ and* any *μ. In particular $\hat{v}$ need not be optimal, and need not even be feasible for* (1)*.*

Corollary 3 is the whole point. The certificate consumes the iterate the solver happened to stop at and the multipliers it happened to report, and returns a valid bound regardless. Computationally it costs one gradient evaluation and a loop over the variables — no second solve.

Remark 4 (Robustness)*.* Because Corollary 3 holds for arbitrary $\lambda \geq 0$ and μ, an error in the sign convention used to recover the multipliers, a stale multiplier, or a multiplier clipped to zero can only make $\hat{q}_s$ smaller. Such an error makes the bound loose, never invalid. Convexity enjoys no such protection; see Section 5.

4 Exactness at a KKT point

The correction term in (4) is not merely small in practice: it vanishes identically at an exact KKT point, so the certificate degrades gracefully and costs nothing when the solve is good.

Proposition 5 (The correction vanishes)**.** *Let $\hat{v}$ satisfy the KKT conditions of* (1) *with multipliers $\lambda \geq 0$ for $g_s \leq 0$, μ for $h_s = 0$, and $z_L, z_U \geq 0$ for the bounds $l - v \leq 0$ and $v - u \leq 0$. Then every term of the sum in* (4) *is zero, and if in addition φ_s is convex then $\hat{q}_s(\hat{v}) = L_s(W_s)$.*

Proof. Stationarity of the full Lagrangian, which is φ_s augmented with the bound terms, reads

$$\nabla \varphi_s(\hat{v}) \;=\; z_L - z_U. \tag{6}$$

Fix a coordinate i and consider the three possible positions of $\hat{v}_i$.

If $l_i < \hat{v}_i < u_i$, complementarity gives $z_{L,i} = z_{U,i} = 0$, so $\partial_i \varphi_s(\hat{v}) = 0$ by (6) and the term is 0 by (5).

If $\hat{v}_i = l_i$, complementarity gives $z_{U,i} = 0$, so $\partial_i \varphi_s(\hat{v}) = z_{L,i} \geq 0$. When the derivative is strictly positive (5) selects $t = l_i = \hat{v}_i$, and when it is zero the term is zero outright; either way the term is 0.

If $\hat{v}_i = u_i$, symmetrically $z_{L,i} = 0$ and $\partial_i \varphi_s(\hat{v}) = -z_{U,i} \leq 0$, so (5) selects $t = u_i = \hat{v}_i$ and the term is 0.

Hence $\hat{q}_s(\hat{v}) = \varphi_s(\hat{v})$. Complementarity $\lambda^\top g_s(\hat{v}) = 0$ and feasibility $h_s(\hat{v}) = 0$ reduce (2) to $\varphi_s(\hat{v}) = f_s(\hat{v}) + W_s^\top x(\hat{v})$. Under convexity the KKT conditions are sufficient for global optimality of (1), so that value is $L_s(W_s)$. ☐

Proposition 5 has a practical corollary worth stating plainly: the correction term *measures its own inexactness*. It is zero when the solve is exact and grows as the solve degrades, so the reported bound loosens smoothly rather than becoming wrong.

5 Canonical form, and where convexity is easy to get backwards

Theorem 2 requires φ_s convex, and by (2) that requires each component of g_s to be convex and each component of h_s to be affine, since $\lambda \geq 0$ but μ is free in sign.

The subtlety is that g_s is the constraint in *canonical* form $g_s(v) \leq 0$, and putting a $\geq$ row into canonical form negates its body. Writing $b(v)$ for the constraint body as the user stated it:

as written	canonical form	requirement on b
$b(v) \le \beta$	$g = b - \beta$	b convex
$b(v) \ge \alpha$	$g = \alpha - b$	b **concave**
$\alpha \le b(v) \le \beta$	both rows	b affine
$b(v) = \beta$	$h = b - \beta$	b affine

So $x^2 \le 4$ is admissible and $x^2 \ge 1$ is not, though both are written with a convex body — and indeed the feasible set of the second is not convex. This is the one place in the construction where an entirely ordinary-looking model silently leaves the hypotheses, and the consequence is not a loose bound but an invalid one. A worked instance: for

$$\min\ x \quad \text{s.t.} \quad x^2 \ge 1, \quad x \in [0, 1.5],$$

whose optimum is 1, taking $\hat{v} = 0.5$ and $\lambda = 1$ in (4) yields $\hat{q} = 1.25 > 1$.

Of the four rows above, three are decidable — affineness is a question about polynomial degree — and the implementation enforces them. Convexity or concavity of a one-sided nonlinear body is not decidable, and remains a user assertion.

6 Aggregating across scenarios

The per-scenario bounds are combined by the usual probability-weighted sum, and the condition under which that sum is itself a bound deserves care.

Proposition 6 (Aggregation)**.** *Let* OPT *be the optimal value of the original stochastic program. If the weights satisfy* $\sum_s p_s W_s = 0$*, then* $\sum_s p_s\, L_s(W_s) \le \mathrm{OPT}$.

Proof. Let x^* together with the scenario recourse decisions be optimal for the original problem, so that $\mathrm{OPT} = \sum_s p_s f_s(v_s^*)$ where v_s^* is the optimal decision in scenario s and $x(v_s^*) = x^*$ for every s by nonanticipativity. Each v_s^* is feasible for subproblem (1), so $L_s(W_s) \le f_s(v_s^*) + W_s^\top x^*$. Multiplying by p_s and summing,

$$\sum_s p_s L_s(W_s) \ \le\ \sum_s p_s f_s(v_s^*) + \Big(\sum_s p_s W_s\Big)^\top x^* \ =\ \mathrm{OPT} + 0. \qquad \square$$

Remark 7 (Weights must share a generation)*.* The hypothesis $\sum_s p_s W_s = 0$ is a *joint* condition on the whole weight vector, and progressive hedging maintains it at each iteration. It is not preserved by mixing: if W_s' takes scenario s's weights from one iteration and scenario t's from another, then in general $\sum_s p_s W_s' \ne 0$, the proof above leaves the residual term $\big(\sum_s p_s W_s'\big)^\top x^*$ in place, and the resulting number is not a bound at all. Nothing downstream can detect this, which is why a scenario whose certificate is unavailable must make the whole expectation unavailable rather than contribute a stale value.

7 Practical consequences

Tightness is governed by the box. By (5) the shortfall of $\hat{q}_s$ below $\varphi_s(\hat{v})$ is

$$\sum_{i=1}^{n} \big|\partial_i \varphi_s(\hat{v})\big| \cdot \big(\text{distance from } \hat{v}_i \text{ to the far end of } [l_i, u_i]\big),$$

so the bound is only as tight as the variable bounds are. Gradient components at a converged solve are small but generically nonzero, so a box of width 10 costs little while a box of width 10^{10} can cost a substantial fraction of the objective. Tightening bounds before certifying — for instance by feasibility-based bounds tightening, which shrinks B without removing feasible points and therefore only raises $\hat{q}_s$ — is the most effective way to improve the bound.

Unbounded variables. If $l_i = -\infty$ and $\partial_i \varphi_s(\hat{v}) > 0$, or $u_i = +\infty$ and $\partial_i \varphi_s(\hat{v}) < 0$, the corresponding term in (5) is $-\infty$ and no finite bound is available. Whether this occurs depends on the model and on how converged the solve is: at an exact KKT point the offending component is zero by Proposition 5, but away from one it generally is not. The implementation reports "no bound" in that case rather than $-\infty$, so that Remark 7 is respected.

Inexact solves. Corollary 3 already covers these, but it is worth spelling out what "inexact" can mean. Problem (1) is convex by hypothesis, so it has no non-global local minima. A solver returning a sub-optimal answer can therefore only mean that it stopped short of converging, leaving an inexact $\hat{v}$ and inexact multipliers. Both are admissible inputs to Theorem 2: the conclusion $\hat{q}_s \le L_s(W_s)$ is unchanged, and the entire effect appears in the correction term of (4), which grows as $\hat{v}$ moves away from a KKT point. By Proposition 5 that term vanishes exactly at a KKT point, so the correction is precisely the price of inexactness, and it measures itself.

Ill-conditioning. The distinct worry is that (1) is badly conditioned, so that the solve is not merely truncated but numerically poor. This affects the tightness of $\hat{q}_s$ and not its validity, for a structural reason: *the certificate is an evaluation, not a solve.* A condition number quantifies how much a linear solve amplifies error — $\kappa(H)$ bounds the relative error in d obtained from $Hd = -g$ — and no step of (5) inverts anything. The computation evaluates φ_s at $\hat{v}$, evaluates $\nabla \varphi_s(\hat{v})$, compares each component with zero, selects an endpoint of the corresponding interval, multiplies and adds. Each is a forward evaluation, and κ has no route in. Correspondingly, the inequality the construction rests on,

$$\varphi_s(v) \;\ge\; \varphi_s(\hat{v}) + \nabla \varphi_s(\hat{v})^\top (v - \hat{v}) \qquad \text{for all } v,$$

is a pointwise consequence of convexity. It holds exactly, at every $\hat{v}$, with no error constant and no dependence on the conditioning of anything. Conditioning governs how loose the plane is away from $\hat{v}$; it cannot change which side of φ_s the plane lies on. What it does change is the size of $\nabla \varphi_s(\hat{v})$ and the quality of the multipliers, and both push $\hat{q}_s$ *down.*

Floating point. Nothing in Theorem 2 requires a tolerance. The arithmetic, being carried out in finite precision, does introduce error, and the previous paragraph says only that this error is not amplified by κ — not that it is absent.

Remark 8 (Rounding). Let u denote the unit roundoff. Evaluating (4) in floating point incurs

$$\left|\mathrm{fl}(\hat{q}_s) - \hat{q}_s\right| \;\lesssim\; u\Big(|f_s(\hat{v})| + \textstyle\sum_i \lambda_i |g_i(\hat{v})| + \sum_j |\mu_j|\,|h_j(\hat{v})| + \sum_i T_i\, d_i\Big),$$

where T_i bounds the intermediate magnitudes arising in $\partial_i \varphi_s(\hat{v})$ and d_i is the distance from $\hat{v}_i$ to the far end of $[l_i, u_i]$. No condition number appears.

Remark 8 is governed by the size of the *summands*, so cancellation matters: multipliers of order 10^8 against an objective of order one put the error floor near 10^{-8} even though $|\hat{q}_s|$ is of order one. The implementation's cushion, reporting $\hat{q}_s - \varepsilon(1 + |\hat{q}_s|)$, is scaled to the size of the *result* instead, and the two agree only on a well-scaled model. The cushion is therefore a numerical safeguard and not part of the argument, and it is exposed as a user-settable quantity rather than fixed.

Note also that the final term of Remark 8 is the one that can raise $\hat{q}_s$. The correction in (5) equals $-\sum_i |\partial_i \varphi_s(\hat{v})|\, d_i$, so a gradient component computed slightly small in magnitude makes it slightly less negative. Remark 4 gives no protection here: it is a statement about the multipliers, and a perturbation of $\nabla\varphi_s$ is not one-directional in the way a perturbation of λ is.

Non-finite values. A solve that diverges can leave NaN in $\hat{v}$ or in the reported multipliers, and an infinite multiplier yields NaN through $\infty \cdot 0$ or an infinite $\varphi_s(\hat{v})$. These propagate silently. The implementation therefore rejects any non-finite $\hat{q}_s$ and reports "no bound", the same disposition it takes for an unbounded direction, so that Remark 7 is again respected. The case that makes this necessary rather than cosmetic is $+\infty$: a consumer keeping the best outer bound seen would discard NaN by accident, since NaN fails every comparison, but would accept $+\infty$ as an improvement.

8 Relation to known results

Nothing in Sections 2–6 is new. This section attributes each result.

Lemma 1 is textbook Lagrangian weak duality, stated for an arbitrary multiplier pair rather than an optimal one.

Theorem 2 is the *Frank–Wolfe duality gap*, sometimes called the Wolfe gap or the linearisation gap. For a convex f on a compact convex D, evaluating the linear minimisation oracle at x gives

$$f(x) + \min_{y \in D} \nabla f(x)^\top (y - x) \;\leq\; \min_D f,$$

which is (4) with $D = B$. The bound is used in the Frank–Wolfe algorithm [1], where the same oracle evaluation that produces it also supplies the search direction; reading it instead as a *certificate* computable at any iterate is the framing in [2]. The only specialisation here is that a box makes the oracle separable and closed-form, so the certificate costs one gradient evaluation and no optimisation at all.

Proposition 6 is the standard argument by which progressive hedging weights yield a lower bound on the optimal value, as in [3].

The combination of progressive hedging with Frank–Wolfe machinery to obtain Lagrangian dual bounds is itself established [4], and mpi-sppy already ships an FWPH cylinder on that basis.

What is specific here is therefore an assembly rather than a result: the multipliers are the ones Ipopt already returns, the linearisation point is the iterate it already stopped at, and the box is the model's own variable bounds — so a valid outer bound is obtained from the solve the spoke was going to perform anyway, with no second optimisation and no assumption that the first one converged. The wider question of extracting valid dual bounds from an inexact oracle is treated at length in the inexact bundle-method literature, which this note does not attempt to survey.

9 Correspondence with the implementation

object here	in the code
φ_s in (2)	`_lagrangian_expression`
$\hat{q}_s$ in (4)	`certified_lower_bound`
decidable hypotheses	`check_model_is_certifiable`
box tightening	`unbounded_variables`
Proposition 6 sum	`SPOpt.Ebound`
Remark 7 safeguard	a `None` bound, propagated collectively
Remark 8 cushion	`eps_rel`, `--ipopt-outer-bound-cushion`
non-finite rejection	`math.isfinite` test on $\hat{q}_s$

The first four live in `mpisppy/utils/dual_certificate.py`; the spoke that drives them is `mpisppy/cylinders/ipopt_outer_bound.py`.

References

[1] Frank, M. and Wolfe, P.: An algorithm for quadratic programming. *Naval Research Logistics Quarterly* **3**(1–2), 95–110 (1956).

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[3] Gade, D., Hackebeil, G., Ryan, S.M., Watson, J.-P., Wets, R.J.-B. and Woodruff, D.L.: Obtaining lower bounds from the progressive hedging algorithm for stochastic mixed-integer programs. *Mathematical Programming* **157**(1), 47–67 (2016).

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